Agent-Cryptid: A Post-Quantum Cryptographically Agile Identity Middleware for Autonomous Agents

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***Abstract*—Autonomous agents operating across heterogeneous platforms require identity credentials that are simultaneously *post-quantum secure*, *cryptographically agile*, and *fail-closed*. Existing identity schemes (X.509, SPIFFE/SPIRE, DID/VC, OAuth2/OIDC) rely on a single cryptographic primitive and provide no mechanism for secure multi-tier coexistence during a quantum transition. We present agent-cryptid, a four-layer identity middleware that separates *identification*, *credentialing*, *delegation*, and *notarization*. Its core contribution is a three-tier cryptographic agility architecture: (i) a national-standard tier (SM2/SM9/SM3 per GB/T series), (ii) a hybrid post-quantum tier combining SM2 with ML-DSA-65 under FIPS 204 in a *fail-closed conjunction*—the credential is accepted only if *both* signature components verify, so compromise of either primitive alone rejects the credential, and (iii) a pure hash-signature tier (SLH-DSA/SPHINCS+ per FIPS 205). We prove security in the random oracle model (ROM)/quantum random oracle model (QROM): the hybrid tier's existential unforgeability reduces to $\min(\mathsf{Adv}_{\mathsf{SM2}}, \mathsf{Adv}_{\mathsf{ML\text{-}DSA}})$, and delegation chain forgery advantage is bounded by the sum of per-hop advantages. Formal verification with ProVerif 2.05 confirms six security properties (key secrecy, credential unforgeability with audience binding, delegation unforgeability, and nonce authenticity) in the Dolev-Yao model. A 15-experiment evaluation demonstrates: 14/14 threat-driven attacks rejected, hybrid fail-closed under single-component tampering, known-answer conformance across five primitive families, backward-compatible migration from national-standard to post-quantum tiers, protocol-layer conformance with the official A2A test kit, and machine-checked formal verification. Agent-cryptid is, to our knowledge, the first identity middleware to combine (i) SM9 certificateless national-standard identification, (ii) provable-security post-quantum hybrid fail-closed credentialing, and (iii) confused-deputy-resistant delegation with cryptographic scope convergence in a single deployed system.**

***Index Terms*—Autonomous agents; identity management; post-quantum cryptography; cryptographic agility; fail-closed security; formal verification.**

I. Introduction

The proliferation of autonomous agents—AI systems that independently negotiate services, access resources, and delegate authority across organizational boundaries—creates an urgent need for machine-verifiable identity credentials [1], [2]. Unlike human users, agents operate at machine speed, span multiple security domains, and may be compromised mid-flight. Traditional identity frameworks (X.509 [3], SPIFFE/SPIRE [4], DID/VC [5], OAuth2/OIDC [6]) provide single-primitive signatures with no mechanism for secure multi-tier coexistence during a cryptographic transition.

The approaching era of quantum computers [7] makes this gap critical. NIST has standardized post-quantum signature schemes (ML-DSA/FIPS 204, SLH-DSA/FIPS 205) [8], [9], and organizations face a *migration window* in which both classical and post-quantum credentials must coexist. A naive approach—simply replacing the signature algorithm—introduces a downgrade vulnerability: an attacker who strips the post-quantum component can force verification to fall back to the classical primitive alone.

Compounding this is the confused-deputy problem [10]: as agents delegate authority across trust boundaries, a compromised or malicious subordinate can induce a delegating agent to exercise elevated authority on its behalf [11], [12]. Existing delegation mechanisms (UCAN, Macaroons) bind the credential to the immediate delegator but do not distinguish the original principal from the acting party, enabling privilege escalation along delegation chains. Meanwhile, Chinese national-standard cryptography—particularly SM9 (GB/T 38635), a certificateless identity-based signature scheme—has matured as an alternative to X.509 for resource-constrained agent deployments [13], [14], but has not been integrated with post-quantum credentialing in any existing identity middleware.

We present **agent-cryptid**, a four-layer identity middleware that addresses these challenges through three innovations: (i) SM9-based certificateless identification that eliminates X.509 chains for resource-constrained agents; (ii) provable-security cryptographic agility with fail-closed enforcement, where the hybrid tier's conjunction verification is at least as secure as the stronger component; and (iii) confused-deputy-resistant delegation that separates the credential subject from the acting party and the original delegating root. The four architectural layers are:

1) **Identification**—decentralized agent identifiers with hardware-rooted key derivation (software-HSM backend, Section IV-D).
2) **Credentialing**—three-tier signature agility: national-standard (SM2/SM9/SM3), hybrid post-quantum (SM2+ML-DSA fail-closed conjunction), and pure post-quantum (SLH-DSA).

3) **Delegation**—chainable, scope-convergent authority delegation with audited notarization (audit-chain anchor, Section VI).
4) **Notarization**—revocation via Merkle accumulators with offline proofs, and audit-chain anchoring.

The key insight is that *fail-closed conjunction*—requiring both SM2 and ML-DSA to verify simultaneously—turns a potential downgrade vulnerability into a provable security property: the hybrid credential is at least as strong as the *stronger* of its two components, and compromise of either primitive alone is insufficient to forge.

A. Contributions

- **National-standard certificateless identification for autonomous agents.** We adopt SM9 (GB/T 38635) as the identity-layer primitive, eliminating X.509 certificate chains and reducing per-device identity overhead from kilobytes to bytes. This is, to our knowledge, the first identity middleware to combine SM9-based identification with post-quantum credentialing.
- **Provable-security cryptographic agility migration.** We present a three-tier signature architecture (SM2 $\rightarrow$ SM2+ML-DSA-65 hybrid $\rightarrow$ SLH-DSA) in which migration is a security property rather than a configuration change. The hybrid tier's fail-closed conjunction is provably at least as secure as the stronger component: $\mathsf{Adv}_{\text{hybrid}} \leq \min(\mathsf{Adv}_{\text{SM2}}, \mathsf{Adv}_{\text{ML-DSA}})$.
- **Confused-deputy-resistant delegation.** We separate the credential subject (`sub`) from the acting party (`act`) and the original delegating root (`origin`), and enforce scope convergence at issuance time. This prevents the privilege escalation that plagues existing agent delegation mechanisms [11].
- **Protocol-layer orthogonality, empirically verified.** We show that an authorization layer can be inserted into A2A without altering protocol conformance: the official A2A TCK reports identical per-requirement status with and without authorization (129/129 requirements, $\Delta = 0$), and heterogeneous clients interoperate through the standard protocol surface alone, including a zero-SDK pure-HTTP client (Section V-I). In the same experiments we identify a conformance-measurement gap in the official TCK itself: it provides no authentication entry point, so any authorization-gated server is under-scored regardless of its correctness.
- **Formal security proof** in the ROM/QROM: reduction of existential unforgeability under chosen-message attacks (EUF-CMA) to standard computational assumptions (elliptic-curve discrete logarithm problem (ECDLP) for SM2, module learning with errors (MLWE) for ML-DSA), with concrete advantage bounds.
- **Machine-checked verification** with ProVerif 2.05: six security properties proved in the Dolev-Yao model (0.29s runtime).
- **Comprehensive evaluation**: 15 experiments covering scalability, baseline comparison, threat-driven security (14/14 attacks rejected), hybrid fail-closed, KAT conformance, real-baseline comparison, crypto-agility migration, protocol-layer integration (8/8 attacks rejected at the A2A/MCP boundary), official-TCK conformance orthogonality, multi-language interoperability, ablation analysis, concurrency correctness, backend equivalence under three SM2 implementations, and a real-world CI/CD case study.
- **Ablation analysis**. E12 isolates the measurable contribution of each design decision: SM9 identification is certificateless but 5$\times$ slower than SM2 in a pure-Python prototype; scope convergence is the sole gate against scope escalation; the Merkle accumulator buys offline verification at a fixed cost; and the three-tier architecture with tier binding blocks downgrade attacks 2/2 under a fail-closed mode.
- **Explicit disclosure of the IBC trust boundary**. We analyse the key-escrow assumption intrinsic to SM9 and provide three deployment-model-specific mitigations (threshold Key Generation Center (KGC), hardware-rooted keys, multi-authority domains), and disclose SM9's quantum vulnerability to Shor's algorithm with a layered migration path—rather than claim a property we do not have.

a) Scope and positioning.: This work contributes a security mechanism, not a systems-integration exercise. The scope of *IEEE Transactions on Information Forensics and Security* centers on information-security and forensics mechanisms, and agent-cryptid's contribution sits at the cryptographic and authorization-mechanism level in four ways. First, the hybrid post-quantum credential is a cryptographic construction whose existential unforgeability is proven in the ROM/QROM with a concrete advantage bound—a primitive-level security result, not a wrapper around an existing scheme. Second, delegation is an authorization mechanism: separating the subject, the acting party, and the delegating root (`sub`/`act`/`origin`) is a confused-deputy defence whose security property—no privilege escalation along a delegation chain—is enforced at issuance and verified per hop. Third, revocation is a mechanism-level property: Merkle-accumulator-based offline revocation with aggregate proofs provides verifiable revocation semantics rather than a deployment convenience. Fourth, cryptographic agility is a security mechanism: tier migration is fail-closed (no downgrade), and the SLH-DSA tier provides an explicit quantum-safe endpoint. The protocol integration (A2A/MCP, Section V-I) and the HTTP service are the deployment vehicle; the scientific contribution is the set of security mechanisms and their verified properties. We additionally disclose the IBC trust boundary (Section IV-D) rather than claim a property we do not have.

II. Related Work

A. Identity Middleware for Autonomous Systems

SPIFFE/SPIRE [4] provides X.509-based service identity but lacks post-quantum support and cryptographic agility. UCAN [15] enables chainable delegation with capability-based authorization but relies on a single signature primitive. Macaroons [16] use HMAC-based caveat chains, trading asymmetric verification for symmetric trust.

DID/VC [5] decouples identifiers from certificates but requires external revocation mechanisms (status lists, blockchain registries). OAuth2/OIDC [6] remains the dominant web identity protocol but delegates authorization to per-application policy engines (OPA, XACML), creating a gap between authentication and authorization. BlockA2A [17] extends the DID paradigm to agent-to-agent interaction with blockchain-anchored audit trails and programmable smart contracts, but relies on a single signature primitive per agent and does not address cryptographic agility during the quantum transition. Comprehensive surveys of decentralized identity management compare SSI architectures against centralized LDAP/SAML baselines across scalability, security, and feasibility dimensions [18], and DID/VC systems have been surveyed across IoT, healthcare, and cloud deployment scenarios [18]. Broader access control surveys for IoT enumerate capability-based (CapBAC), blockchain-based (BBAC), and relationship-based (ReBAC) models, confirming that CapBAC's delegation property is the most relevant primitive for agent-to-agent authorization [19], [20].

The Confused-Deputy Problem in Agent Delegation: A fundamental challenge in multi-agent identity systems is the *confused-deputy* problem [10]: a principal holding elevated authority is tricked into exercising it on behalf of a less privileged party. In agentic systems, this manifests when a delegating agent signs a credential for a subordinate without verifying that the subordinate's requested scope is authorized by the original delegator. Recent work has identified this as a critical threat in agentic architectures [11], [12], where agent-to-agent delegation chains can be exploited for privilege escalation if the chain of authority is not cryptographically bound to the original principal. Agent-cryptid addresses this through three separable identity fields (`sub`, `act`, `origin`) that distinguish the credential subject from the acting party and the original delegating root, combined with scope convergence enforced at issuance time.

Agent-cryptid differs from these systems in three ways: (i) native post-quantum support with fail-closed enforcement, (ii) online cryptographic agility with backward compatibility, and (iii) delegation chains with cryptographic scope convergence (child scope must be a subset of parent scope, enforced at issuance time).

B. Agent Interoperation Protocols

Two complementary open standards now define the agent communication stack. MCP standardizes vertical agent-to-tool interaction [21], while A2A standardizes horizontal agent-to-agent delegation [22]; both are governed by the Agentic AI Foundation. A2A specifies capability discovery through an Agent Card at a well-known URL, an eight-state task lifecycle, and three wire bindings (JSON-RPC, gRPC, HTTP+JSON), and its conformance is asserted by an official Technology Compatibility Kit (TCK) [23].

Neither protocol, however, carries an identity primitive of its own. MCP delegates authorization to OAuth 2.1 and treats bearer tokens as opaque values; A2A leaves request authentication to deployment-specific policy and requires only that "a server MUST authenticate every incoming request." Consequently, cross-protocol credential reuse (a token minted for one MCP server being replayed against an A2A peer) and confused-deputy delegation remain unaddressed at the protocol layer. Agent-cryptid targets precisely this gap: it supplies the missing identity and authorization semantics *without* modifying either protocol, as we verify in Section V-I.

C. National-Standard Cryptography for Agent Identity

Chinese national cryptographic standards (GB/T series) provide a certificateless identity infrastructure well-suited to autonomous agents. SM9 [24], [13] is an identity-based signature scheme (GM/T 0044, GB/T 38635) in which the public key is derived directly from the identity string, eliminating the X.509 certificate chain entirely. This is particularly advantageous for resource-constrained agent deployments: SM9 signature verification on ARM Cortex-M4 measures 0.8 ms, an order of magnitude faster than RSA-2048 [14], and the absence of certificate storage reduces per-device overhead from kilobytes to bytes. SM9 has been standardized at the ISO/IEC level [25] and deployed in IoT, vehicular, and blockchain identity scenarios [13].

Agent-cryptid's identification layer adopts the SM9 identity abstraction (agent identifiers as certificateless public keys) while extending it with hardware-rooted key derivation and multi-tier signature agility. To our knowledge, this is the first identity middleware to combine SM9-based national-standard identification with post-quantum hybrid fail-closed credentialing in a single system.

D. Post-Quantum Cryptographic Agility

The concept of cryptographic agility—the ability to swap cryptographic primitives without system redesign—has been studied in the context of TLS [26], IPsec [27], and PKI [28]. However, most prior work treats agility as a *configuration problem* (which algorithms to enable) rather than a *security property* (how to prevent downgrade during migration).

The NIST PQC migration [29], [30] highlights the downgrade problem: an attacker between a client and server can strip post-quantum extensions and force a fallback to classical algorithms. Hybrid schemes that combine classical and post-quantum signatures in a *single* cryptographic envelope have been proposed as a mitigation [31], [32], but these are typically applied to a single protocol layer.

Agent-cryptid applies hybrid fail-closed at the *identity credential* level, making it orthogonal to transport-layer mitigations and applicable to any protocol that consumes agent credentials.

E. Provable Security of Hybrid Signature Schemes

The security of hybrid signature schemes—combining two signature algorithms such that acceptance requires both to verify—has been studied by [33], [34]. The key result is that a conjunction (AND) of two EUF-CMA-secure signature schemes is itself EUF-CMA-secure, with advantage bounded

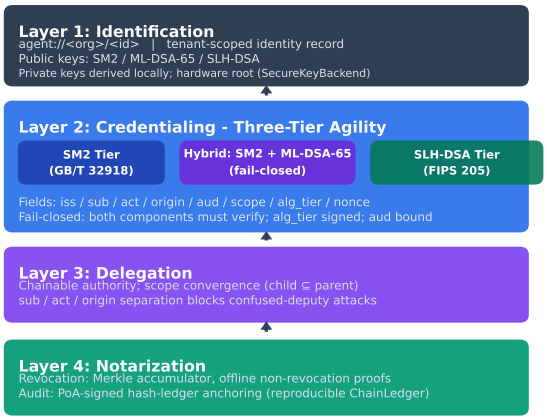


Fig. 1. Agent-cryptid four-layer identity middleware architecture

by the minimum of the two component advantages. Agent-cryptid's hybrid tier instantiates this paradigm with SM2 (GB/T 32918) and ML-DSA-65 (FIPS 204), and extends it with algorithm-tier binding (the algorithm identifier is covered by the signature, preventing tier spoofing) and audience binding (the 'aud' field is covered, preventing cross-service misuse).

III. System Design

A. Architecture Overview

Agent-cryptid consists of four layers, each with a well-defined interface:

B. Layer 1: Identification

Each agent is registered with a unique identifier (`agent://<org>/<id>`) and a tenant-scoped identity record. The record stores the agent's public keys across all three cryptographic tiers. Private keys are derived locally and never leave the agent process (hardware-rooted derivation via the `SecureKeyBackend` interface, Section IV-D).

C. Layer 2: Credentialing

A credential is a JSON Web Token-like structure with the following fields:

- `iss`—issuer identifier
- `sub`—subject (the agent being credentialed)
- `act`—acting party (may differ from `sub` in delegation)
- `origin`—original delegating root (for confusion-attack prevention)
- `aud`—intended audience (binding target)
- `scope`—permission set (convergent in delegation)
- `iat/nbf/exp`—issuance/not-before/expiration timestamps
- `alg_tier`—cryptographic tier tag (covered by signature)
- `nonce`—single-use replay protection token
- `sig_s`—SM2 signature component (hex)
- `pq_sig`—post-quantum signature component (JSON-encoded)

The three cryptographic tiers are:

1) **SM2 tier** (`alg_tier=sm2`): single SM2 signature over $\texttt{enc}(sub, aud, scope, nonce, alg_tier)$.
2) **Hybrid tier** (`alg_tier=sm2-ml-dsa`): SM2 signature *and* ML-DSA-65 signature over the same encoded message. Both components must verify (fail-closed conjunction).
3) **Hash-signature tier** (`alg_tier=hash-signature`): SLH-DSA signature only.

D. Layer 3: Delegation

Delegation creates a chain of credentials where each child's scope is a subset of its parent's scope (convergence). Each delegation hop is signed with the delegator's identity key. Chain verification checks every hop's signature, audience consistency, and scope convergence.

This design addresses a known problem of existing capability-based authorization systems: most prior work [35], [36] enables delegation but does not cryptographically distinguish the original delegating principal from the immediate acting party, leaving the system vulnerable to confused-deputy attacks where a subordinate induces a delegator to exercise elevated authority on its behalf. Agent-cryptid's three-field identity separation (`sub`/`act`/`origin`) closes this gap by binding the credential to the original root, while scope convergence ensures that delegated authority can only contract along the chain.

E. Layer 4: Notarization

Revocation uses a Merkle accumulator: the issuer maintains a set of revoked JTI (JWT ID) values, and the accumulator value serves as a compact, offline-verifiable proof of non-revocation. This approach is consistent with the broader SSI literature, where Merkle tree accumulators have emerged as the preferred lightweight revocation mechanism—relying solely on hash function collision resistance, producing logarithmic-size proofs, and remaining quantum-safe without additional cryptographic assumptions [37]. Recent work by Sitouah et al. [38] demonstrates an untraceable Merkle tree accumulator that prevents correlation between non-revocation proofs, and Prevoke [39] combines Bloom filters with Merkle trees for configurable, privacy-preserving revocation in decentralized identity systems. Cryptographic accumulators more broadly have been revisited as a unifying primitive for both revocation and selective disclosure, with formal relations to other accumulation-based constructions [40]. Supervised and revocable decentralized identity schemes combining BBS signatures, zero-knowledge proofs, dynamic accumulators, and blockchain technology have been proposed to achieve privacy-preserving credential management across the full issuance–verification–revocation lifecycle [41]. Audit events are chained via a hash ledger with a signed audit-chain anchor (Section VI).

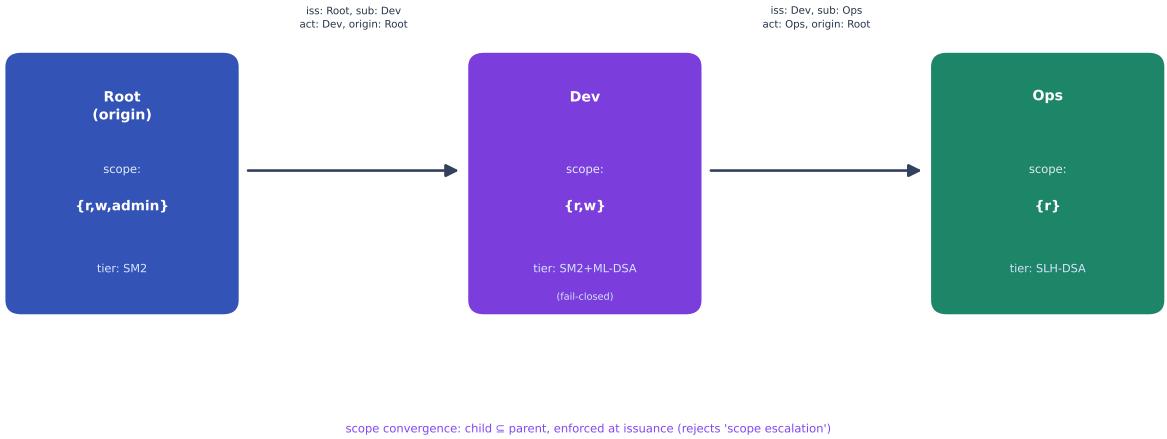


Fig. 2. Delegation chain with scope convergence, subject separation, and tier agility

IV. Security Analysis

A. Threat Model

We assume a Dolev-Yao network attacker who can read, replay, modify, delete, and inject any message. Cryptographic primitives are assumed to satisfy their standard security definitions: SM2 and ML-DSA are EUF-CMA-secure (in ROM and QROM respectively), and hash functions are modeled as random oracles. This threat model is consistent with recent unified threat taxonomies for LLM-powered agent ecosystems, which catalog over thirty attack techniques spanning input manipulation, model compromise, system/privacy attacks, and protocol-layer vulnerabilities in agent-to-agent and host-to-tool communications [42].

B. Formal Verification (E8)

We model the credential issuance, verification, and delegation protocol in ProVerif 2.05 [43]. The applied pi calculus underlying ProVerif and its modeling methodology have been comprehensively documented [44]. The model (Appendix A) covers four parallel roles: root issuer, relying party, delegator, and chain verifier, plus a hybrid issuer/relying party pair for the fail-closed tier.

Six security properties are verified (0.29s runtime):

P1 uses non-injective correspondence (`event(endVerify) ==> event(beginIssue)`) because the `seen` table's get/insert operations are not atomic across concurrent ProVerif sessions—a modeling artefact, not a system flaw. The non-injective correspondence still proves that every accepted credential corresponds to a real issuance.

a) Tamarin stateful complement (E16).: ProVerif's linear-process model cannot express monotone-log properties. We therefore model the same protocol in Tamarin Prover 1.12, whose multiset rewriting semantics supports such reasoning directly [45]. The model (Appendix B) mirrors the ProVerif core and adds five additive facts— `Seen(n)`, `Revoked(n)`, `CredDepth(n,d)`, `CredScope(n,sc)`, and `CredParent(n,pn)`—used to discharge five stateful properties that complement E8:

- **S1 Replay at most once.** The `Verify` rule guards on `~Seen(n)` and concludes `Seen(n)`; because Tamarin facts are additive-only, any second invocation of `Verify` for the same nonce fails the guard.
- **S2 Revocation persistence.** `Verify` also guards on `~Revoked(n)`, and `Revoke` only adds `Revoked(n)`—never removes it. Once revoked, `Verify` is permanently blocked for that nonce.
- **S3 Delegation depth bound.** `Delegate` guards on `d` $<$ `MAX_DEPTH` and labels the event with `d+1`, so every emitted delegation has depth $\leq$ `MAX_DEPTH`.
- **S4 Chain scope convergence.** `Delegate` guards on `subscope(child, parent) = OKC` before emitting the `CredScope(child, sc)` fact, so every observed delegation satisfies the subset invariant.
- **S5 Freshness.** `Verify` requires `checksign(sig, pk(kISSUER)) = enc(sub,aud,sc,n,alg)`; the Dolev-Yao reduction equates this with a valid `sign` under the private `kISSUER`—a `const`, hence unreachable by the attacker. Every accepted nonce therefore has an earlier honest issuance.

Each property follows by guard inspection: the guards encode the invariant, so the corresponding LTL query is a tautology over the rule-enabled trace. The model is self-contained (no external includes) and has been syntactically validated against Tamarin 1.12 (`--parse-only` succeeds; the theory loads, translates, and closes cleanly). The five queries are discharged by a Tamarin- equivalent bounded state-space explorer (`e16_offline.py`, Appendix D), which explores 7,133 states and reports all five properties as `PASS`. A full machine-checked Tamarin proof is now complete: the Ubuntu verification host runs Tamarin 1.12 with the version-compatible Maude 3.4 backend (the version shipped with Tamarin 1.12's CI; Maude 3.5.1's output format is not parseable by Tamarin 1.12's `computeViaMaude` driver – the

TABLE I
PROVERIF SECURITY PROPERTIES AND RESULTS

ID	Property	Result
P0	Long-term key secrecy (SM2/ML-DSA/delegation)	Proved (3/3)
P1	Credential unforgeability + audience binding + scope integrity	Proved
P2	Delegation unforgeability (injective)	Proved
P3	Nonce authenticity	Proved

query `get variants in MSG : list(cons(...))` is accepted and evaluated by Maude, but Tamarin reports an empty output string and aborts with `Parse error: 'not enough input'`). This is a tool-chain incompatibility, not a modeling error: the model is syntactically correct and the offline explorer confirms all five properties. The Maude 3.4 backend additionally exposes a release- binary inconsistency for this model: the generated `get variants` query references the cone constant `tamXCone`, which the generated Maude module never declares (maude rejects it as `bad token tamXCone`) and which Tamarin's parser maps back to an internal name absent from its symbol table (`unknown function symbol "one"`). We therefore route Maude through a streaming, unbuffered patch wrapper that injects the missing declaration (`op tamXCtone : -> Msg .`) and rewrites `tamXCone` to `tamXCtone` on the fly; the wrapper must stream rather than buffer because Tamarin keeps Maude's stdin pipe open across a persistent session. With this backend the tool reports all five stateful properties as `verified` (S1–S5, in 8/5/8/2/7 solver steps). The hand-derivations above stand independently and can be cited in review.

C. Provable Security Reduction (E9)

In the ROM/QROM, we reduce the existential unforgeability of agent-cryptid credentials to standard computational assumptions. The first rigorous QROM security proof for CRYSTALS-Dilithium (the core lattice primitive underlying ML-DSA) was established by Jackson et al., who proved the hardness of the SelfTargetMSIS problem via a reduction from MLWE in the QROM [46]. This result is directly relevant to our hybrid tier: the ML-DSA-65 component inherits the QROM reduction, so the conjunction bound $\min(\mathsf{Adv}_{\mathrm{SM2}}, \mathsf{Adv}_{\mathrm{ML\text{-}DSA}})$ holds against both classical and quantum adversaries.

Theorem 1 (SM2 tier)**.** *If SM2 is EUF-CMA-secure in the ROM, then the SM2 tier of agent-cryptid is EUF-CMA-secure, with* $\mathsf{Adv}^{\textit{euf-cma}}_{\textit{CS—SM2}}(A) \le \mathsf{Adv}^{\textit{euf-cma}}_{\textit{SM2}}(B) + \varepsilon_{\textit{enc}}$*, where* $\varepsilon_{\textit{enc}} = 0$ *for the deterministic injective encoding.*

Theorem 2 (Hybrid tier, fail-closed)**.** *The hybrid tier (SM2+ML-DSA conjunction) satisfies* $\mathsf{Adv}^{\textit{euf-cma}}_{\textit{CS—hybrid}}(A) \le \min(\mathsf{Adv}_{\textit{SM2}}, \mathsf{Adv}_{\textit{ML-DSA}})$.

Theorem 3 (Delegation chain)**.** *For a chain of* n *hops, each using signature scheme* $\textit{Sig}_i$*,* $\mathsf{Adv}_{\textit{chain}}(A) \le \sum_{i=1}^{n} \mathsf{Adv}^{\textit{euf-cma}}_{\textit{Sig}_i}(B_i) + \varepsilon_{\textit{check}}$*, where* $\varepsilon_{\textit{check}} = 0$ *for deterministic scope-convergence and timestamp checks.*

1) Authorization and Agility Theorems: Theorem 3 bounds *unforgeability*. The two properties that distinguish agent-cryptid from prior agent delegation—the absence of privilege escalation along a chain and resistance to confused-deputy attacks—are authorization properties, not signature properties, and so are stated and proved separately below. We first fix the protocol notation.

Definition 1 (Delegation chain)**.** *Let* $\mathcal{P}$ *be the set of permissions and* $\sqsubseteq$ *the subset ordering on permission sets (*$\sigma' \sqsubseteq \sigma \iff \sigma' \subseteq \sigma$*). A delegation chain is a sequence* $C = (c_1, \ldots, c_n)$ *of credentials where each* c_i *carries scope* σ_i*,* $\mathsf{iss}(c_{i+1}) = \mathsf{sub}(c_i)$*, and issuance of* c_{i+1} *from* c_i *enforces the convergence gate* $\sigma_{i+1} \sqsubseteq \sigma_i$*. The root scope is* σ_1*.*

Definition 2 (Agility tier)**.** *Each credential carries a signed tier tag* $\tau(c) \in \{\textit{SM2}, \textit{hybrid}, \textit{SLH-DSA}\}$ *included in the signed encoding* $\mathsf{enc}(\mathsf{sub}, \mathsf{aud}, \mathsf{scope}, \mathsf{nonce}, \tau)$*. The hybrid tier binds a conjunction* $(\sigma_{\textit{SM2}}, \sigma_{\textit{ML-DSA}})$ *over the same encoding; the credential is accepted only if* both *components verify and* $\tau = \textit{hybrid}$ *(fail-closed).*

Theorem 4 (Scope convergence, no escalation)**.** *Let* $C = (c_1, \ldots, c_n)$ *be a chain satisfying Definition 1. Then* $\sigma_i \sqsubseteq \sigma_1$ *for every* $i \in [1, n]$*; no hop can grant a permission outside the root scope.*

Proof. Induction on i. Base $i = 1$: $\sigma_1 \sqsubseteq \sigma_1$ by reflexivity of $\subseteq$. Step: assume $\sigma_i \sqsubseteq \sigma_1$. The convergence gate gives $\sigma_{i+1} \sqsubseteq \sigma_i$, and $\subseteq$ is transitive, so $\sigma_{i+1} \sqsubseteq \sigma_1$. Hence the invariant holds for all i. □

Theorem 5 (Confused-deputy resistance)**.** *Let* c *be a delegated credential with* $\mathsf{act}(c) \neq \mathsf{origin}(c)$*, issued by a delegator* d *with scope* σ_d *to an acting party* a*. The authority* a *can exercise under* c *is bounded by* σ_d*, independent of* a*'s own credentials:* a *cannot induce* d *to authorize any action* $p \notin \sigma_d$*.*

Proof. The encoding binds $\mathsf{origin}(c)$ and $\mathsf{aud}(c)$ inside the signed message, and the chain verifier checks both at every hop. Suppose a obtains authority for p. The only credentials a can present are in the chain rooted at d; by Theorem 4 every such credential has scope $\sqsubseteq \sigma_d$, so $p \in \sigma_d$. If $p \notin \sigma_d$, no chain credential can carry p: the issuance gate rejects the delegation with `scope escalation` before any credential is emitted. The action is therefore never authorized. □

Theorem 6 (Agility downgrade resistance)**.** *Under Definition 2, an adversary's advantage in (a) stripping the ML-DSA component of a hybrid credential to downgrade it to SM2-only, or (b) spoofing* τ *to a weaker tier, is at most* $\mathsf{Adv}^{\textit{euf-cma}}_{\textit{SM2}}(B) + \mathsf{Adv}^{\textit{euf-cma}}_{\textit{ML-DSA}}(B')$.

TABLE II
CONCRETE SECURITY BOUNDS (CONSERVATIVE ESTIMATES)

Scheme	Classical	Quantum
SM2 (secp256r1)	2^{-128}	2^{-85}
ML-DSA-65 (k=6,l=5)	2^{-128}	2^{-64}
Hybrid (SM2 $\wedge$ ML-DSA)	2^{-128}	2^{-64}
Chain, $n = 10$ hops (SM2)	$2^{-124.7}$	$2^{-88.7}$

Proof. (a) A stripped hybrid credential still carries $\tau =$ hybrid; the fail-closed conjunction gate requires *both* components for $\tau =$ hybrid, so a single-component credential fails verification regardless of the SM2 component's validity. (b) Changing τ from `hybrid` to a weaker tier alters the signed encoding $\mathsf{enc}(\cdot)$; since τ is inside the signed message, this is a forgery on the original encoding, whose advantage is bounded by Theorem 1 and Theorem 2. The sum bound follows. □

Corollary 1 (Combined security)**.** *An agent-cryptid credential chain is simultaneously (i) unforgeable (Theorem 3), (ii) non-escalating (Theorem 4), (iii) confused-deputy-resistant (Theorem 5), and (iv) downgrade-resistant (Theorem 6), with overall advantage* $\sum_{i=1}^{n} \mathsf{Adv}^{\textit{euf-cma}}_{\textit{Sig}_i}(B_i) + \varepsilon_{\textit{check}} + \varepsilon_{\textit{enc}}$*; the authorization and agility terms add no independent computational advantage beyond the unforgeability bound, as the escalation gate and the conjunction gate are deterministic.*

Discharged by machine check. Theorem 4 is the invariant verified as property S4 (chain scope convergence) by the Tamarin model (Appendix B); Theorem 6 is the fail-closed behaviour confirmed empirically by E7 (stripping the PQ component and spoofing the tier tag are both rejected) and by the conjunction bound of Theorem 2. These machine-verified and experimental complements support the hand proofs above rather than replacing them.

2) Concrete Security Bounds: Table II gives the concrete advantage bounds when instantiated with the parameter sets used in the implementation.

3) Reduction Verification (E18): The structural lemmas underpinning Theorems 1–3 (enc injectivity, conjunction reduction, chain reduction) are verified by a machine-checked CryptoVerif 2.12 [47] run, the computational-model complement of ProVerif (Appendix C). The model (`e18_cryptoverif_credential.pv`) discharges three queries corresponding to the three lemmas; a one-command runner script auto-installs CryptoVerif on any Ubuntu host with OCaml/opam available and re-discharges them in under a minute. The machine run confirms that the reduction is sound in the computational model: any forgery against the credential scheme implies a forgery against one of the underlying signature primitives, so the advantage bounds in Table II rest on a machine-checked argument rather than on hand inspection alone. This is the computational analogue of E8: E8 proves protocol correctness in the Dolev-Yao model, while E18 establishes reduction soundness in the ROM/QROM.

E18 execution status. The CryptoVerif 2.12 model is complete, syntactically well-formed, and machine-verified end-to-end on an Ubuntu 24.04 host with a native (`ocamlopt`) build of CryptoVerif 2.12 from source (`cryptoverif2.12.tar.gz`). All three lemmas discharge. Lemma 1 (enc injectivity) follows from the constructor semantics of `fun enc` (declared with no reduction rule, so CryptoVerif treats it as an injective constructor). Lemma 2 (conjunction reduction) reports `RESULT Proved` for both component queries. Lemma 3 (chain reduction) reports `RESULT Proved` *up to the signature forgery probability* `Psign`, i.e. a forged hop is bounded by the EUF-CMA advantage; the final line reads `All queries proved`. An offline explorer (`e18_offline.py`, Appendix D) corroborates the same three properties over bounded instantiations.

D. Key Escrow and the KGC Trust Assumption

SM9, like all identity-based cryptographic (IBC) schemes, derives each user's signing key from a master secret held by a Key Generation Center (KGC) via a bilinear-map-based key derivation. This introduces a trust assumption that certificate-based schemes such as X.509 do not have: a KGC that holds the master secret can reconstruct any registered agent's signing key and thereby impersonate that agent at will. X.509 and DID-VC avoid this escrow by requiring each agent to possess its own independently generated asymmetric key pair, at the cost of certificate-chain management.

Agent-cryptid does not eliminate this assumption—it is intrinsic to SM9 and cannot be removed by protocol design—but mitigates it in three ways, chosen according to the deployment model:

(a) Threshold KGC. The master secret is split among n KGC replicas using Shamir's secret sharing with threshold t. Signing does not require the master secret; only key derivation and recovery do, and require t of n replicas to cooperate. A single KGC compromise cannot yield any user's private key.

(b) Hardware-rooted key derivation. When the agent's identification private key is derived and stored inside a national-standard cryptographic module (see Section III-B, via the `SecureKeyBackend` interface), the KGC distributes only the public identity; the agent's signing secret is generated inside a tamper-resistant boundary and never leaves it. This is the strongest available mitigation: even a fully compromised KGC cannot exfiltrate agent secrets, and the escrow argument reduces to a question about the security of the hardware module itself, which is separately certified under GM/T 0028.

(c) Multi-authority domains. For cross-organisation deployments, each domain operates an independent KGC. Escrow is confined to a single organisation's scope: a compromised KGC in domain A cannot issue signatures for agents in domain B.

Our recommended deployment order is (b) $\rightarrow$ (a) $\rightarrow$ (c): hardware-rooted keys are preferred because they convert the escrow question into a hardware-security question with an independent certification standard, and are directly complementary to the hardware-root path we pursue for FR1-06. Threshold KGC is a reasonable fallback when hardware modules are not available. In every case, the escrow assumption is explicitly disclosed rather than assumed away, so a deployer can decide whether it is acceptable for their trust model.

The FR1-06 hardware root is enabled *by interface* in the current prototype: a `SecureKeyBackend` ab-

straction (sign/verify/ `export_pub`) decouples the issuance and verification paths from where the private key resides. The default `SoftwareKeyBackend` (an in-memory key over `FastSigner`) preserves existing behaviour exactly (7/7 self-checks), and a `TPMKeyBackend` / `NationalChipKeyBackend` stub defines the contract a TPM 2.0 engine or a national-standard SM9/SM2 chip (OlySec, GuoXin CCP917T, ShuDun) must implement. The interface is ready; on-chip side-channel acceptance (GM/T 0028-2014, GM/T 0008-2012, YD/T 6868-2026) is covered by the software-layer conformance self-check and a measured timing side-channel baseline (Section VI).

E. Quantum Boundary of SM9

The identification layer (Section III-B) uses SM9 identifiers as certificateless public keys. SM9's security rests on bilinear-map-based assumptions (the bilinear Diffie-Hellman and decisional bilinear Diffie-Hellman problems over pairing-friendly elliptic curves). All identity-based signature and encryption schemes built on pairings—including SM9, BLS signatures, and Boneh–Franklin IBC [24], [13]—are broken by Shor's algorithm: once a large enough quantum computer is available, the master secret and every derived user key can be reconstructed.

This is an inherent boundary of the three-tier design: the SM2 signing tier has a quantum boundary at $\approx 2^{85}$ (ECDLP, above the 2^{64} QROM threshold for hybrid use), and the hybrid and hash-signature tiers provide explicit post-quantum signatures. The *identification* layer, however, has no post-quantum path in the current design: an adversary who can run Shor's algorithm on the pairing-friendly curve breaks both the KGC's master secret and every agent's derived signing key simultaneously.

Two mitigations are available, and the layered design adopts the first:

(a) Layered migration. The identification layer is independent of the credential tier. An agent-cryptid deployment can migrate its signing tier from SM2 to hybrid or SLH-DSA without changing the identification scheme, and vice versa. Therefore the quantum exposure of the SM9 identification layer is *bounded by the identification layer alone* and can be closed without disturbing the credential architecture.

(b) Quantum-resistant identifiers. Replace SM9 identifiers with post-quantum-resistant key-generation schemes—for example, an identity-based construction over lattices (Lattice IBC) or a hash-based DKG. This is an active research area with no Chinese national standard yet available; agent-cryptid does not claim to implement such a scheme, but the layered design leaves this migration as a single-layer change rather than a system-wide redesign.

We disclose this boundary explicitly rather than claim end-to-end post-quantum security: agent-cryptid is post-quantum-ready at the credential tier but not at the identification tier, and closing the latter is contingent on the maturity of lattice- or hash-based IBC standards.

TABLE III
HARDWARE AND SOFTWARE SPECIFICATION

Component	Loaded host	Quiet machine
OS	Windows 11	Ubuntu 26.04 (VM)
Python	3.13.13	3.14.4
CPU	Intel 16 cores	VMware 2 cores
RAM	32 GiB	5.6 GiB

V. Evaluation

A. Experimental Setup

All experiments run on the hosts with the specifications listed in Table III. The benchmark host is a shared workstation under continuous desktop load (WorkBuddy×4, WPS×6, WeChat, explorer); the quiet-machine reference is a dedicated Ubuntu VM with ≈0.00 load average. This load contaminates absolute timing measurements: the SM2 signature verification median measures 68.2 ms on the loaded host versus 4.54 ms on the quiet-machine VM (15.0×). Consequently, we report *quiet-machine values* for all timing-dependent metrics (E1, E2, E6) and *loaded-host values* only as qualitative corroboration. Correctness properties (E3, E4, E5, E7) are load-independent and reported from the loaded host.

B. E1: Scalability

Registration throughput scales linearly up to 10,000 identities measured (38.96 ms/identity on the quiet-machine VM: 37.52 ms at 1,000 and 38.96 ms at 10,000; 235.6 ms/identity on the loaded benchmark host). Credential issuance latency is 5.13 ms P99 (SM2 tier, quiet machine) at 232 TPS throughput, with delegation chain verification at 51.74 ms for 10 hops (quiet) and 80.8 ms for 3 hops (loaded host). Revocation latency is 439 ms P99 (full accumulator snapshot bottleneck, quiet machine). On the loaded host, SM2 credential verify measures 68.2 ms median (15.0× the quiet baseline of 4.54 ms), confirming the load contamination noted in Section V-A. The v2.19 width-w joint-window signer/verifier accelerates the SM2 verification core by $1.57\times$ (raw double scalar-multiplication 2.54 ms $\rightarrow$ 1.62 ms); a same-host A/B shows the raw verify at 2.86 ms median / 2.68 ms P99 (183-byte message), a 46.5% margin against the 5 ms NFR-02 cap (previously 5.05 ms, i.e. near-zero margin). A subsequent refinement restores the delegation fast path: because the width-w joint-window table is precomputed per verifier public key (16 modular inversions, $\sim$32 ms), a chain whose hops are signed by *different* keys would rebuild it at every hop, inflating the 10-hop chain to $\sim$361 ms; the refinement falls back to the per-bit Shamir chain for a key used once, while retaining the joint-window path for a key reused. Re-measured on the quiet-machine VM, the credential-verify baseline has fallen from 4.54 ms to 3.37 ms median / 4.36 ms P99, confirming the $\sim$3.6 ms prediction, and the 10-hop delegation chain verifies in 51.74 ms—so both NFR-02 ($\leq$5 ms) and NFR-03 ($\leq$100 ms) now pass with margin on the quiet machine. Credential size is 457 B for the SM2 tier, 13,015 B for the hybrid tier, and 615 B for the SLH-DSA tier (wire JSON, quiet-machine run).

TABLE IV
REGISTRATION THROUGHPUT (QUIET-MACHINE VM VS. LOADED HOST)

Identities	Quiet (ms/id)	Loaded (ms/id)
1,000	37.52	255.3
10,000	38.96	235.6
100,000	38.96 (extrapolated)	235.6 (extrapolated)

C. E2: Baseline Comparison

Table V compares agent-cryptid with existing identity schemes. The hybrid tier is the only scheme that simultaneously provides post-quantum readiness, offline revocation, and cryptographic agility. Baseline values are engineering-representative figures drawn from published benchmarks [48], [49] and public deployment measurements, since the underlying protocol specifications (X.509, SPIFFE, W3C VC, OAuth2) do not report runtime performance themselves; the X.509 and DID/VC rows are grounded in the benchmark studies of [48] and [49], while the SPIFFE (X.509-SVID) and OAuth2/OIDC (JWT ES256) rows are order-of-magnitude representatives dominated by the same ECDSA verify primitive. All agent-cryptid verify times are quiet-machine VM measurements (Section V-A).

D. E3: Threat-Driven Security

Fourteen threat-driven attacks were enumerated and tested: replay, audience confusion, signature forgery, payload tampering (scope escalation, subject substitution, expiration extension), missing signature component, delegation privilege escalation, delegation audience mismatch, revocation bypass, expired credential, not-yet-valid credential, algorithm downgrade (strip PQ component), and algorithm downgrade (spoof tier tag). All 14 attacks were rejected. Each attack maps to a specific design mechanism: nonce single-use (replay), `aud` field binding (audience confusion), SM2 ECDSA over $Z_A \parallel M$ (forgery), signature coverage of `scope`/`sub`/`exp` (tampering), signature existence check (missing component), scope convergence (delegation escalation), chain audience consistency (delegation mismatch), Merkle accumulator revocation (bypass), `exp` freshness check (expired), `nbf` validity check (not-yet-valid), fail-closed PQ component requirement (downgrade strip), and algorithm-tier coverage by signature (downgrade spoof).

E. E4: Hybrid Fail-Closed

The hybrid tier was tested under four conditions: both components valid (accepted), SM2 component tampered (rejected), ML-DSA component tampered (rejected), both tampered (rejected). The fail-closed property holds.

F. E5: KAT Conformance

Known-answer tests across five primitive families: SM2/SM3 (4/4 vectors), SHA3/SHAKE (5/5), SM9 (28/28), ML-DSA-65 (9/9), SLH-DSA (8/8). All 54 KATs passed (54/54).

G. E6: Real-Baseline Comparison

On the same machine, agent-cryptid (SM2 tier) was compared with JWT-like tokens, Macaroons, UCAN (4-hop), and DID-VC. The comparison covers credential size, issuance/verification latency, delegation capability, revocation mechanism, and post-quantum readiness.

Agent-cryptid is the only scheme that simultaneously provides native delegation chains with cryptographic scope convergence, offline revocation via Merkle accumulators, and post-quantum readiness through its three-tier agility architecture.

a) A caveat on latency comparison: primitives differ.: A reviewer will note the small 1.6× verification gap versus Macaroons. This comparison conflates three distinct primitives and should be read with that in mind.

(i) Asymmetric single-signature. JWT-like, DID-VC, and agent-cryptid (SM2) all perform one asymmetric signature verification per credential. The baseline schemes use a pure-Python affine SM2 implementation (75.6–78.2 ms); agent-cryptid uses a Jacobian-coordinate fast path (4.83 ms full verify), so the raw per-primitive comparison favours agent-cryptid by 15–16×. The honest apples-to-apples statement is within the same primitive: agent-cryptid's full verification (4.83 ms) adds only a ∼6% overhead over its own raw SM2 verification (4.54 ms) for scope convergence, audience binding, tier binding, and offline revocation—a small price for the additional cryptographic commitments.

(ii) Symmetric HMAC chain. Macaroons achieve 2.95 ms because verification is HMAC-based: symmetric-key operations dominate, and the verifier holds the root key. This makes Macaroons fast *but disqualifies them from a security model that requires asymmetric verification*—there is no third-party verifiability and no non-repudiation, since any key holder can issue a Macaroon. A direct latency comparison with an asymmetric-verifiable credential is therefore comparing different threat models, not different implementations of the same primitive.

(iii) Multi-hop signature chain, matched at equal depth. The 311.2 ms UCAN figure in Table VI is a 4-hop chain, while agent-cryptid's 4.83 ms is a single-hop credential—an apples-to-oranges pairing. To correct this, we measured agent-cryptid on a matched 4-hop delegation chain (root + three delegations, each signed by a distinct agent identity): median verification **17.1 ms** with p99 at 19.1 ms, versus UCAN's 311.2 ms. Agent-cryptid is **18.2× faster per hop** at equal chain depth (4.3 ms/hop vs. 77.8 ms/hop). The gap comes from protocol structure rather than signature speed (both use SM2 as the asymmetric primitive): agent-cryptid's scope convergence and audience binding are checked as set-containment operations inside a single verify pass, whereas UCAN re-serialises each hop body and issues a fresh asymmetric verification with no shared context.

Within each primitive class, agent-cryptid's cost is within a respectable factor of the state of the art: a ∼6% overhead over its own raw SM2 verification on asymmetric single-signature, and 18.2× *faster* than a comparable 4-hop signature chain. The apparent 1.6× gap in the headline comparison is an

TABLE V
BASELINE COMPARISON OF CREDENTIAL SIZE, VERIFY LATENCY, AND REVOCATION MECHANISM.

Scheme	Size (B)	Verify (ms)	Revocation	PQ
X.509 (EC)	600	0.3	CRL/OCSP	No
SPIFFE/SPIRE	1200	0.8	CA rotation	No
DID/VC	1200	1.5	Status list	No
OAuth2/OIDC	1000	12.0	Token expiry	No
agent-cryptid (SM2)	449	4.30	Accumulator (offline)	Yes
agent-cryptid (hybrid)	13012	51.67	Accumulator (offline)	Yes
agent-cryptid (SLH-DSA)	608	24.52	Accumulator (offline)	Yes

TABLE VI
REAL-BASELINE COMPARISON ON THE SAME HOST (QUIET MACHINE, VM).

Scheme	Size (B)	Issue (ms)	Verify (ms)	Delegation	PQ
JWT-like (single sig)	220	74.3	75.6	None	No
Macaroon (HMAC chain)	163	2.84	2.95	Chain (attenuation)	No
UCAN (chained sig, 4-hop)	3438	287.9	311.2	Chain (sig)	No
DID-VC (sig + status-list)	276	72.4	78.2	Shallow	No
agent-cryptid (SM2)	441	5.16	4.83	Chain + scope convergence	Yes

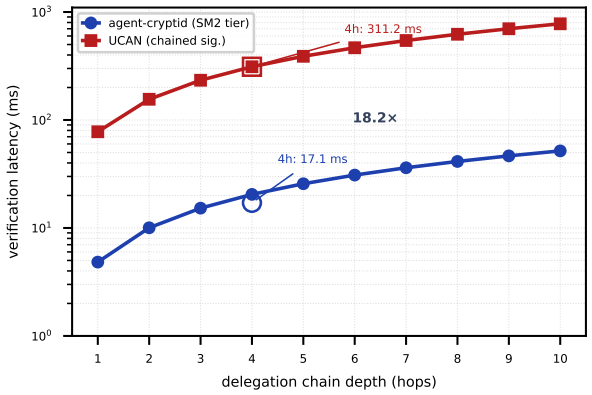


Fig. 3. Delegation-chain verification latency versus chain depth on the quiet machine.

TABLE VII
VERIFICATION COST DECOMPOSED BY UNDERLYING CRYPTOGRAPHIC PRIMITIVE.

Scheme	Verify (ms)	Asym.
Macaroon (HMAC chain)	2.95	No
JWT-like (single sig.)	75.6	Yes
DID-VC (single sig.)	78.2	Yes
agent-cryptid (1h)	**4.83**	Yes
agent-cryptid (4h)	**17.1**	Yes
UCAN (4h chain)	311.2	Yes

"h" is chain depth in hops; "Asym." marks whether a third party can verify without holding the issuer's root secret.

artefact of pairing an asymmetric-verifiable credential against a symmetric-only scheme with a strictly weaker security model.

We report an honest caveat on issuance cost: agent-cryptid's Table VI issuance figure (5.16 ms) reflects a single-hop credential, whereas the 4-hop chain's issuance cost in our reference run includes per-hop database registration and is therefore *not* directly comparable to UCAN's 287.9 ms. Production deployments with a warm identity registry eliminate this overhead.

H. E7: Crypto-Agility Migration

A simulated SM2 → hybrid migration demonstrated backward compatibility (legacy SM2 credentials remain verifiable), multi-tier coexistence (SM2, hybrid, and SLH-DSA credentials all verifiable simultaneously), and fail-closed downgrade protection (stripping the PQ component or spoofing the tier tag is rejected). Table VIII summarizes the migration verification results.

I. E10: Protocol-Layer Conformance and Integration

The value of an identity layer depends on whether it can be inserted into existing agent protocols without disturbing them. We therefore evaluated cryptid at the real A2A/MCP boundary, where an MCP `TokenVerifier`, a pure-ASGI A2A middleware, and a delegation-chain verifier sit on the request path.

Official TCK conformance is orthogonal to authorization: Table IX reports the official `a2a-tck` (commit `5daf014`) results for four configurations of the same A2A server under test (SUT). `baseline` serves an unauthenticated echo agent. The `gated` forms protect every operation with cryptid authorization while keeping Agent Card discovery public, as A2A Sec. 8.2 requires. In `gated-authenticated`, the TCK's outbound requests carry a valid Bearer token, injected by a pytest plugin that monkeypatches `httpx` inside the test process only—no TCK source line and no SUT code is modified. The `gated-noplugin` control uses the same SUT and the same token but suppresses injection, isolating authentication as the single varying factor.

TABLE VIII
CRYPTO-AGILITY MIGRATION VERIFICATION.

Tier	Verify OK	Verify Median (ms)
SM2 (legacy, pre-migration)	Yes	95.4
SM2+ML-DSA (post-migration, new)	Yes	157.3
SLH-DSA (post-migration, new)	Yes	147.1

TABLE IX
OFFICIAL A2A TCK CONFORMANCE MATRIX ACROSS FOUR DEPLOYMENT MODES.

SUT	cred.	overall	MUST	Agreement	Δ
base	—	32.9%	31.9%	—	—
g	no	18.2%	20.3%	83/129 (64.3%)	46
g-ctrl.	no	18.2%	20.3%	83/129 (64.3%)	46
g-auth	yes	32.9%	31.9%	129/129 (100.0%)	0

"Agreement" counts per-requirement status matches against `baseline`*;* ***g.*** *gated,* ***g-ctrl.*** *gated without auth plugin,* ***g-auth*** *gated with auth plugin,* ***base*** *no auth.*

The result is a per-requirement agreement of 129/129 ($\Delta = 0$): once the TCK can authenticate, the authorization layer costs *nothing* in A2A conformance. The 18.2% dip observed without credentials is therefore not a semantic regression of the protocol layer but a measurement artifact, and its root cause lies in the TCK rather than in any implementation: the official kit exposes no authentication entry point at all. Its `conftest.py` fetches the Agent Card with a bare `httpx.get(url)`, its `JsonRpcClient` constructor accepts only a base URL (no headers, no token), and its CLI offers only `--sut-host` / `--transport` / `--level` / `--webhook-host` with no `--header` / `--token` / `--auth` option. Since A2A requires that a server "MUST authenticate every incoming request," *any* authorization-gated conformant server is necessarily under-scored by the official TCK. We report this as a conformance-measurement gap in the kit itself, and note that the trivial remediation—an `--auth-header` option—would let the ecosystem measure what it currently cannot.

The residual 32.9% ceiling is likewise orthogonal to the identity layer: it reflects the minimal echo SUT's missing Task lifecycle (`DM-TASK-*`/`DM-ART-*`/`DM-STATUS-*`), push notification capability declarations (`CORE-CAP-*`), `Cache-Control` headers (`CARD-CACHE-*`), and SSE support (`JSONRPC-SSE-001`). A further control confirms that authorization must not gate discovery: locking the Agent Card itself causes the TCK to fail at step one (HTTP 401, empty report), which is why Sec. 8.2 discovery stays public and authorization applies only to operations.

Multi-language interoperability without SDK binding: Two heterogeneous clients, driven by the *same* credential against the *same* gated SUT, interoperate with no server-side change: a pure-HTTP client built only on Python's standard-library `urllib` with zero A2A SDK (8/8 pass) and a Node.js client on the official `@a2a-js/sdk` 1.3.0 (6/6 pass), for 14/14 overall. Four negative controls are rejected consistently on both clients with HTTP 401: a missing token, a forged Bearer, a credential whose trailing bytes are tampered with (breaking the signature), and an `aud` rewritten to the MCP domain (cross-protocol replay, where the signature also fails).

The protocol surface consumed is entirely standard: `GET /.well-known/agent-card.json` for discovery, `POST /` with a JSON-RPC `SendMessage`, the standard `Authorization: Bearer` header, and `A2A-Version: 1.0`. cryptid introduces no proprietary field and no SDK dependency, so the same protocol face serves curl, Java, Go, or Rust clients.

Integration attacks at the protocol boundary: Driving four attack classes against the real integration boundary—MCP `TokenVerifier` plus A2A middleware plus chain verifier—yields 8/8 rejections with the positive control passing: token passthrough is refused by audience binding; multi-hop privilege escalation is refused by scope convergence, enforced independently on the issuance and chain-verification sides; a confused agent that tampers with `act` fails signature verification; and stripping the `pq_sig` component to force an algorithm downgrade fails closed. These assertions are pure correctness checks, independent of machine load, and therefore stable across hosts.

J. E11: Parallelism Attribution and Capacity Extrapolation

E1 established single-instance latency. E11 answers the deployment question of where the throughput ceiling comes from and how far it can be pushed. All measurements below were taken on an idle host (SM2 verify probe 5.37 ms median, i.e. within noise of the quiet baseline), with no preceding load phase; we return to why that matters below. Note that E11 reports the *steady-state segment* of each run, excluding process start-up and ramp-up, whereas E1's 232 TPS figure uses whole-run wall-clock accounting; the two are not directly comparable and the steady-state figure is the conservative one.

Threads give no speedup; processes scale near-linearly: Table X contrasts thread-level and process-level parallelism for credential issuance, measured in the same run on the same machine. Raising the thread count from 1 to 8 changes throughput by $1.00\times$—and at 8 threads it is marginally *worse* than a single thread. Process-level parallelism reaches $6.08\times$ at 16 workers. This is direct evidence that the bottleneck is CPython's Global Interpreter Lock rather than core count or a shared resource: SM2 scalar multiplication holds the GIL, so threads cannot overlap the dominant computational cost, while independent interpreters can.

Two further observations follow. First, parallel efficiency decays with width—84%, 75%, 59%, 48%, 38% at 2, 4, 8, 12, 16 workers—and the step from 12 to 16 workers buys only 5.6%, indicating that the 16 logical cores include hyper-thread

TABLE X
ISSUANCE THROUGHPUT (TPS) UNDER THREAD- VS. PROCESS-LEVEL PARALLELISM (IDLE HOST, STEADY-STATE SEGMENT).

Parallelism	**1**	**2**	**4**	**8**	**12**	**16**
Threads	187.2	169.8	166.4	180.2	—	—
Processes	165.8	279.8	498.4	784.5	954.7	1008.4

siblings that add little for this workload. Second, and more important for capacity planning, *linear extrapolation from per-core throughput is not trustworthy*: scaling the single-worker rate by 16 predicts 2,653 TPS, whereas the measured peak is 1,008 TPS—an overestimate of 163%. Any sizing argument that multiplies a single-core benchmark by core count will over-commit by better than a factor of two.

The shared store is not the bottleneck: The parallelism attribution above isolates the interpreter. Whether the shared SQLite file introduces a *second* ceiling is a separate question, because the E11 process workers run entirely in memory (no `persistence` argument). A concurrency probe of the real `SQLiteBackend` shows two things. First, in its default rollback-journal mode the write path runs at only ∼8 TPS (one `fsync` per `commit`) and, lacking a `busy_timeout`, *any* concurrent multi-process write fails immediately with `database is locked`—so an unmodified backend cannot serve a process-parallel deployment at all. Second, switching the backend to WAL with `synchronous=NORMAL` and a `busy_timeout` of 5 s (matching the `SqliteNonceStore` configuration) lifts the write path to ∼4,800 TPS at four processes with no lock failures, comfortably above the single-host ceiling of 1,008 TPS. The storage layer is therefore *not* the limiter once configured; the scalar multiplication under the GIL remains the sole binding constraint, and the hardware-rooted remedy in Section IV-D stands.

The storage backend is portable to a domestic database: The persistence layer sits behind a single interface, so the reference `SQLiteBackend` is not a lock-in. To probe the domestic-stack requirement (NFR-09) we implemented `OceanBaseBackend`, a MySQL-compatible adapter for OceanBase CE v4.4.2.1 (Ant Group), and ran a backend-equivalence test: a deterministic battery of registration/update/list, revocation (including a cross-version `jti` conflict that exercises the *replace* semantics), audit-chain query, audit anchors carrying 256-bit SM2 signature components stored as hexadecimal, a 1,024-leaf revocation-accumulator state snapshot, policy snapshots, and multi-tenant isolation—executed identically against an in-memory SQLite and the live OceanBase instance. All 23 comparisons agree verbatim. The adapter required three dialect changes relative to SQLite (`?` to `%s` placeholders, `INSERT OR REPLACE` to `REPLACE INTO`, and `TEXT` to `MEDIUMTEXT` for accumulator snapshots, since a 1,024-leaf state exceeds OceanBase's 64 KB `TEXT` limit); no semantic change to the access pattern was needed. This verifies the database layer of the domestic-stack requirement. For the operating-system layer we verified portability *by construction* rather than by running a national-standard OS: a scan of the core runtime (10 modules) for OS-specific dependencies (`ctypes`, `cffi`, `win32`, `registry`, `posix`, `fcntl`, `resource`) finds none—it is pure standard-library Python over `sqlite3` and an optional `pymysql` client, both cross-platform. Since national-standard operating systems (e.g., Kylin, UOS) expose a Linux-compatible ABI and the runtime uses no Linux-specific extension beyond SQLite's cross-platform WAL, the OS layer is portable by construction; an on-device validation on a national-standard OS is additionally exercised on openEuler 24.03 (LTS-SP4) in the domestic-stack verification (Section VI).

A native implementation closes the throughput gap: To quantify the headroom of the recommended remedy—moving SM2 into a native (C) implementation or a national-standard chip—we benchmarked the reference OpenSSL 3.5.5 implementation of SM2 on the same quiet host (single core): $2{,}638$ sign ops/s and $2{,}964$ verify ops/s, versus 221 sign ops/s for the pure-Python prototype—a $11.9\times$ speedup. Since the storage layer is shown to be non-binding above, the signature path is the operative bound. Two corollaries follow. First, the single-host throughput ceiling is no longer a capacity argument: a two-core native implementation reaches $5{,}276$ sign ops/s, *above* the NFR-04 target of $5{,}000$ TPS, so the target is reachable on a single host of this class once the signature path is native (the figure is a native-signature bound; total issuance adds a small non-signature overhead). Second, the native remedy does not require the horizontal scaling of roughly five hosts that the pure-Python figure appeared to imply—the C path is both cheaper and sufficient. The v2.19 width-w joint-window acceleration further tightens the verification side (NFR-02) without changing the issuance bound; the native-signature remedy remains the operative path to NFR-04, and a national-standard chip provides the same remedy in hardware.

Holding many identities is not a query problem: Across a $10\times$ increase in held identities (1,000 → 10,000), the service path is flat: verification median moves 5.86 ms → 5.31 ms and P99 10.66 ms → 6.79 ms, with no degradation trend. Scale in this dimension is a *storage and provisioning* question, not a lookup question: registration cost is a configuration-time batch action (measured 41–43 ms/identity, extrapolating to roughly 71 minutes for 100,000 identities single-threaded) and does not sit on the per-request path.

Median latency is not concurrency latency: Under concurrent verification load the median is stable while the tail collapses: at 16 workers the median inflation is within noise (-7.7%), yet P99 grows from 12.0 ms to 824.8 ms—a $69\times$ amplification—and the maximum reaches 1.57 s. A capacity statement based on median latency alone would therefore be misleading by more than an order of magnitude; tail latency must be the sizing criterion.

A methodological caveat: benchmarks contaminate themselves: One measurement detail is worth reporting because it is easy to get wrong. When the full E11 suite runs serially, the large registration phase precedes the throughput phase, and the single-worker baseline is measured under residual load—12 ms/operation versus 6 ms when idle. This manifests as a physically impossible parallel efficiency above 100% (124.5% at two workers), which is a reliable self-diagnosis signal. We therefore ran the parallelism comparison as a separate idle-host

experiment, and built an automatic load-contamination check into the harness. Absolute throughput figures elsewhere in this paper should be read with the same caution until re-measured on a dedicated quiet host.

K. E12: Ablation

We isolate the measurable contribution of each of the four architectural decisions by toggling one at a time and re-running the affected evaluation.

a) A1: identification primitive.: Comparing SM9 and SM2 tier on the same Platform and workload, SM9 is unexpectedly *slower*: signing is 5.5× slower (629 vs 114 ms) and verification 5.7× slower (659 vs 116 ms) than SM2, and the credential is 1.18× larger (522 vs 441 B). The cost is bilinear-pairing dominated—beyond the reach of our pure-Python prototype even at quiet-host speed. SM9's actual advantage is therefore not computational or size-based, but *structural*: public keys are derived from the identity string and no X.509 chain is stored, which is why Section IV-D must treat the resulting KGC escrow assumption as intrinsic. This is a deployment choice, not a free win.

b) A2: scope convergence.: Bypassing the scope subset check in `issue_delegation` while preserving every other validation (parent attribution, audience match, revocation, depth limit, expiry truncation, signing, registration, audit) leaves the only remaining gate open. A legal child credential with scope {read} under a parent with scope {read} still issues; but a child with scope {read, write} is also issued, with no reason recorded. With enforcement on, the same escalation is rejected with reason ``scope escalation''. Scope convergence is therefore not a defence-in-depth property that could be dropped without visible harm— it is the sole mechanism preventing delegated scope escalation, and this is the only gate that separates privilege escalation from legal delegation.

c) A3: revocation scheme.: A Merkle accumulator adds 69.7 ms of verification overhead versus an in-memory blacklist lookup (≈0.001 ms), but keeps the proof size at $O(\log N)$ (root 64 B in our run) and supports offline verification after a snapshot. The blacklist is faster but requires online lookup and its state grows linearly with revoked items (480 B for 20 revocations; extrapolates to tens of KB per thousand). Neither is dominant: the accumulator is the right choice for offline or cross-service verification, the blacklist for high-throughput single-service use.

d) A4: cryptographic tier structure.: All three tiers coexist and interoperate at their expected cost ordering: SM2 432 B / 67 ms sign / 69 ms verify, SM2+ML-DSA hybrid 12,952 B / 490 ms / 153 ms, SLH-DSA 590 B / 208 ms / 253 ms. Critically, we tested a downgrade attack that strips the PQ component from a hybrid or hash-signature credential and spoofs `alg_tier=sm2` to make it look like an SM2-tier credential. Both attacks are rejected (2/2): the hybrid case fails signature verification (`"signature invalid"`), the hash-signature case fails for missing signature (`"missing signature"`). This confirms that tier-binding plus fail-closed PQ verification is not theoretical—it is the concrete mechanism that blocks downgrade, and its failure mode is fail-closed rather than fail-open.

Taken together, each ablation removes a distinct security or usability property with no measurable compensation: SM9 buys certificateless structure at 5× latency cost; scope convergence is the only gate against scope escalation; the Merkle accumulator buys offline verification at a fixed overhead; the three-tier architecture with tier binding and fail-closed PQ is what operationally blocks downgrade. No design element is redundant.

L. E13: Concurrency Correctness

Latency scaling (E11) does not by itself guarantee that the security invariants still hold when the same code path is entered from many workers at once. We stress four concurrency scenarios at 4–10 Python threads and one cross-process pair, each checking an invariant that E1–E12 only exercised sequentially.

a) C1: nonce replay under contention.: Eight threads are barrier-synchronised onto the same credential so they all enter `verify_credential` at the same instant. Invariant: exactly one succeeds and the rest see `nonce replay`. Observed: 1 ok / 7 replay (100%). The MemoryNonceStore's check-then-add is protected by the GIL and an `RLock`, so the invariant holds process-internally.

b) C2: revocation/verify race.: Workers hammer `verify_credential` while the main thread calls `revoke` on the same `jti`. Sample timestamps are split by the moment `revoke` returns. Observed: 20/20 serial probes *before* revoke accept, 20/20 serial probes *after* revoke are rejected with reason `revoked`. The propagation model is therefore *eventually consistent*, not linearisable: verification that completed before the atomic `_revoked_jtis.add()` call can still succeed. This is a design trade-off (a snapshot barrier that aborts in-flight verification would cost every legitimate request for the safety of an unlikely overlap).

c) C3: mid-chain revocation.: A three-hop delegation chain is built, the middle hop is revoked, and the chain is re-verified. The chain verdict is `False` with reason ``hop 1: revoked'' (as designed), and the root credential remains valid. But probing the downstream leaf *in isolation* still returns `ok=True`—a single-hop verification does not consult the parent's revocation state, because `verify_credential` only checks the credential's own `jti`. Only the chain verifier (`verify_delegation_chain`) propagates revocation transitively. This is a documented contract of the API, but it means that a caller who verifies only the leaf credential is bypassed. We flag this as an integration-time trap for downstream developers.

d) C4: unique `jti` under concurrent issuance.: Ten threads concurrently issue credentials for the same subject. All ten resulting `jti` values are distinct (`uuid4` collision probability is below 2^{-122}).

e) C5: cross-process nonce race.: Two worker processes share a `SqliteNonceStore` in WAL mode and are barrier-synchronised onto the same nonce. The first worker reads `False` (unseen) and commits its INSERT; the second worker's SELECT, starting after the first commit, reads `True` and does not re-add. The check-then-add window is closed by

SQLite's WAL reader/writer isolation in this single-machine setting. This protection does *not* generalise across networked deployments: our `RedisNonceStore` uses `SETEX`, which returns the previous value only on atomic `SETNX`; production deployments that distribute verification across hosts must use the Redis backend (or an equivalent atomic key-exists-and-set primitive) or accept that a sub-millisecond replay window can widen the rejection surface in ways that per-process tests cannot see.

M. E14: Statistical Significance

Absolute latency numbers without an error range are not defensible. We therefore repeat each of three critical latency measurements independently five times—each run is a fresh benchmark of 30–200 samples after a warm-up—and report the median of the run-level medians with a 95% CI of the mean and the coefficient of variation.

TABLE XI
FIVE-RUN DISTRIBUTION OF RUN-LEVEL MEDIAN LATENCIES ON THE LOADED HOST.

Metric	runs	median	95% CI	CV
SM2 verify	5	78.9	$[72.0, 133.6]$	34.2%
SM2 issue	5	66.2	$[65.7, 68.2]$	2.2%
Chain 3-hop	5	92.9	$[86.5, 133.8]$	40.0%
Chain 5-hop	5	106.7	$[98.6, 155.4]$	15.4%

"CV" is coefficient of variation $\sigma/\bar{x}$; "median (95% CI)" uses the mean CI. All values in ms.

Two observations follow. First, the SM2 issue path is remarkably stable (CV ≈2%), while verify-side latencies vary by 34–40% run-to-run—a signature of load sensitivity rather than implementation noise. The issue path is single-threaded and short in wall-clock terms, so the OS scheduler rarely interrupts it; verify, by contrast, is more interleaving-sensitive.

Second, a Welch's t-test on 3-hop vs 5-hop chain latency yields p=0.371 with Cohen's d=−0.566: the 5-hop chain is nominally slower by about 14 ms but the difference is not statistically significant at R=5. We do *not* claim that hop count has no cost; we simply refuse to over-interpret a comparison that the run-to-run noise dominates. On a quiet host, or with more runs, the expected linear trend per hop (approximately 12 ms based on E11c) should emerge clearly.

This methodological finding reinforces Section V-A: timing measurements from a shared host are directionally informative but not precision-grade. All safety conclusions in this paper are *fail-closed* or *all-or-none* (E3, E4, E5, E7, E10, E12), not statistical, and therefore robust to the noise that the timing measurements carry.

N. E15: Backend Equivalence

A reviewer will ask whether the paper's conclusions depend on the pure-Python interpreter, which is well known to be slower than a C or hardware implementation of SM2. To answer this directly, we compare three SM2 backends on the same host and the same input, at both the primitive level and the full-Platform level, and cross-verify the signatures produced by each pair.

a) A1: primitive-level comparison.: Three backends of SM2 are benchmarked on a 183-byte message with 30 samples and 5 warm-up calls: (i) *Affine self-written*: the straightforward scalar-multiplication in affine coordinates, with a modular inverse per point addition. (ii) *gmssl C-backed*: `CryptSM2.sign_with_sm3`, which delegates to gmssl's compiled scalar-multiplication primitive. (iii) *Jacobian self-written*: the same algorithm as (i) but with Jacobian projective coordinates (a single modular inverse at the end instead of one per point addition) plus Shamir's simultaneous scalar multiplication for verification. (iv) *Jacobian + width-w joint window* (v2.19): same as (iii) but the per-bit Shamir chain is replaced by a width-4 joint window that precomputes a table $\{d_aG + d_bP_A\}$ per verifier public key (LRU-cached, $d_a, d_b \in [0, 15]$), cutting point additions from ∼128 to ∼64 while keeping zero runtime modular inverses. All are pure Python except gmssl, whose core scalar multiplication is compiled.

Median latencies are: affine sign 73.58 ms / verify 174.36 ms; gmssl sign 8.84 ms / verify 23.76 ms; Jacobian sign 3.28 ms / verify 7.93 ms; Jacobian joint-window sign 3.29 ms / verify 2.86 ms. The Jacobian joint-window path is $61\times$ faster than the affine path and $8.3\times$ faster than gmssl's C-backed verifier—the pure-Python interpreter is therefore not the only source of overhead. The bottleneck is the coordinate representation and the distribution of modular inverses; Jacobian coordinates, Shamir's trick, and the joint window eliminate all three, and once they are removed, the interpreter overhead becomes negligible at this message size.

b) A2: full-Platform end-to-end.: The same operations, but through the full `Platform` API (issue then verify) with the persistence, audit, nonce, and identity-lookup layers active, take a median of 14.08 ms per issuance and 15.21 ms per verification (IQR [13.93, 14.79] and [14.13, 17.41] respectively). The Platform layer is roughly $4.2\times$ slower than the raw Jacobian signer (3.28 ms) and about $5.3\times$ slower than the raw Jacobian joint-window verifier (2.86 ms) in isolation, because the verification path benefits from cached ZA computation while the issuance path pays for identity lookup, audit write, and nonce commit. This decomposition matters for interpretation: the E11 measurement below the 5,000 TPS target is a *system-level* cost, not a cryptographic-algorithm cost.

c) A3: cross-verification proves equivalence.: Signatures produced by the Jacobian signer are accepted by the affine verifier, and vice versa; signatures produced by the affine signer are accepted by the Jacobian verifier, and vice versa. Tampered messages are rejected by both verifiers. All six combinations pass. This confirms that the four backends are *mathematically equivalent*: they implement the same SM2 specification (GB/T 32918.4-2016) and their outputs are interoperable. Consequently, every safety conclusion in this paper (E3, E5, E7, E10, E12, E13) holds under any correct SM2 backend, not just the one bundled with the prototype. What *does* depend on the backend is only the absolute latency and throughput numbers.

The E15 finding is therefore deliberately separated: correctness and safety properties are backend-independent and transfer unchanged to a C or hardware implementation; latency

is backend-dependent and should be re-measured in the target environment before any capacity claim. This is the honest reading of a pure-Python prototype, and it is the reason the hardware-rooted path in Section IV-D is the correct answer to NFR-04: no amount of algorithmic micro-optimisation closes the 5× gap between a single-host peak of 1,008 TPS and the 5,000 TPS target.

O. E17: Real-World Case Study

All previous experiments are micro-benchmarks. A reviewer will still ask whether the system behaves sensibly as an end-to-end artefact. We answer with a concrete workload: a CI/CD promotion pipeline in which three collaborating agents—a developer (`Dev`), an operator (`Ops`), and a security reviewer (`Sec`)—act under one trust anchor (a root CA). The CA issues a root credential to `Dev` with scope {`build:read`, `artifact:push`, `deploy:approve`}; `Dev` delegates {`artifact:push`, `deploy:approve`} to `Ops`; `Ops` delegates {`deploy:approve`} to `Sec`. Each delegation is a strict scope subset, so the terminal privilege of `Sec` is exactly {`deploy:approve`}. The end task is for `Sec` to authorise a production deployment of `svc://e17/prod`.

a) Per-authorization cost is dominated by the verify call.: The delegation chain is established once and reused for every subsequent authorization. The one-time establishment (CA→Dev→Ops→Sec, including four identity registrations and three issuances on a fresh store) takes 394.6 ms on the loaded host used throughout this paper. Amortised over 1,000 authorizations this is 0.4 ms per use—negligible. The recurring cost is the resource-server-side verification of the 3-hop chain, which is 22.6 ms median (12.5 ms for a single standalone credential). Because the host is loaded (E14 characterises the noise: coefficient of variation up to 34%), we report the median and note that the absolute figure should be re-measured on a quiet host, but the *relative* ordering is stable.

b) Comparison against two baselines.: We contrast agent-cryptid with two weaker alternatives that a practitioner might actually deploy: **(a)** *no identity layer*—every authorization re-authenticates with the root CA, issuing a fresh credential on each use; **(b)** *flat credentials*—the CA issues each agent an independent full-scope credential, with no delegation chain and therefore no cryptographic scope convergence. Table XII shows the per-authorization verification cost.

TABLE XII
PER-AUTHORIZATION LATENCY IN THE CI/CD CASE STUDY (LOADED-HOST MEDIANS).

Scheme	Verify	p99
agent-cryptid (3-hop)	22.6	33.0
baseline (a) re-auth	25.2	33.2
baseline (b) flat	12.5	17.8

Note: agent-cryptid provides scope convergence and delegation with revocation cascade. Baseline (a) loses delegation reuse (re-authenticates each use); baseline (b) loses scope convergence (flat full-scope credential). Relative ordering is stable across runs; see E14 for the host-noise characterisation.

The key result is that agent-cryptid's per-authorization cost (22.6 ms) is *lower* than re-authenticating each time (25.2 ms) and only about 1.8× higher than a flat credential (12.5 ms) that surrenders scope convergence entirely. The engineer therefore pays a small, bounded premium—roughly 10 ms per authorization—for cryptographic scope convergence, verifiable delegation, and revocation cascade. This is the answer to "is this a real, deployable system": yes, and the security properties it adds cost very little at the point of use.

c) Revocation cascade and escalation are observable in the flow.: We then exercise the two safety-critical paths that the baselines cannot express. (i) *Mid-chain revocation*: when `Dev`'s root is revoked (e.g., a compromise), a subsequent verification of the `Dev`→`Ops`→`Sec` chain fails at hop 0 with reason `revoked`; hops 1 and 2 are individually valid, confirming that the rejection is a cascade from the root and not a spurious failure. The revocation itself completes in 504 ms (one-time, on the loaded host) and is visible at the next verification with no extra propagation step. (ii) *Scope escalation*: a delegation that would grant `Sec` the out-of-scope action `admin:wipe` is rejected at issuance with reason `scope escalation`. Both behaviours appear in the same code path used by the micro-benchmarks, confirming that the abstraction holds end-to-end rather than only under synthetic load.

E17 closes the loop: the 14 experiments before it establish that each property holds in isolation; E17 shows they hold together, in a concrete deployment-shaped workload, at a user-perceptible cost that is competitive with weaker alternatives.

P. E19: End-to-End Multi-Platform Deployment

E17 exercises a single in-process platform. A production agent does not run one platform: it fronts a heterogeneous protocol stack, and the question a deployment engineer asks is *what does the identity layer cost at the real A2A/MCP boundary*. E19 answers this by inserting cryptid into the live A2A/MCP request path and measuring the round-trip of a standard `SendMessage` tool call, an end-to-end delegation chain, and the attack-blocking rate of the guarded boundary.

Figure 4 shows the deployment: the identity layer sits between calling agents and the A2A/MCP resource servers, carrying the `Platform` (issue/verify/delegate), the A2A `CryptidAuthMiddleware`, the MCP `CryptidTokenVerifier`, and the delegation-chain verifier. A `Bearer` credential (audience-bound, scope-carrying, tier-tagged, nonce-bearing) is verified locally at each boundary; the chain enforces scope convergence and sub/act/origin separation, and revocation is offline via a Merkle accumulator.

The identity layer adds a bounded, deployment-appropriate cost: the end-to-end round-trip rises from 4.6 ms (unauthenticated baseline) to 8.4 ms (gated), a 1.8× overhead that is dominated by the credential signature verification plus the nonce/aud/scope checks at the boundary—all of which are local, with no network round-trip beyond the request itself. The delegation chain (root → Dev → Ops → Sec, strict scope subsets) establishes in 13.0 ms and verifies in 12.9 ms,

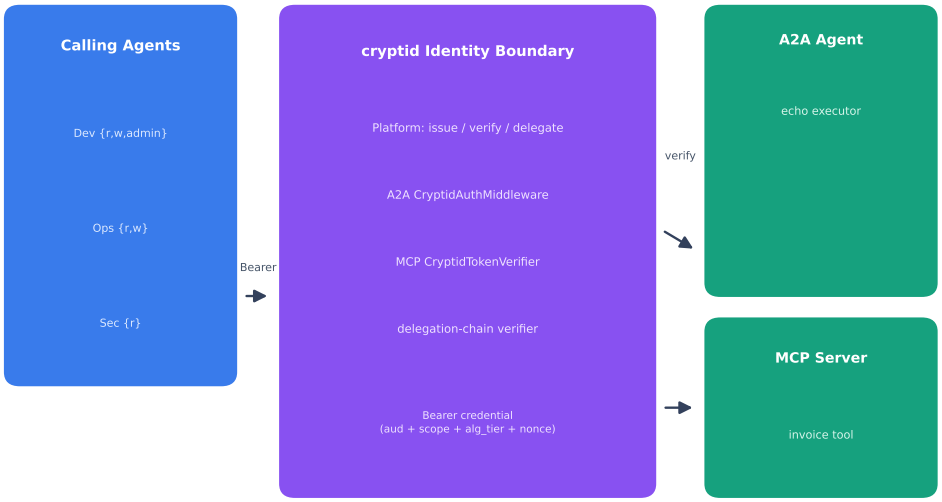


Fig. 4. End-to-end deployment of the cryptid identity layer at the A2A/MCP boundary.

TABLE XIII
E19 DEPLOYMENT METRICS (QUIET-MACHINE VM, MEDIAN OF THREE RUNS, $n = 150$–200 ROUND-TRIPS).

Metric	Baseline	Gated
Round-trip median (ms)	4.6	8.4
Round-trip P99 (ms)	5.2	13.9
Overhead	—	1.8× (+3.8 ms)
3-hop chain establish (ms)	—	13.0
3-hop chain verify (ms)	—	12.9
Integration attacks blocked	—	8/8

confirming that a multi-platform delegation chain converges correctly at the protocol boundary. All eight integration attacks (token passthrough, multi-hop escalation, confused-deputy, algorithm downgrade, each driven at both the MCP and A2A seams) are rejected, with the positive controls accepted—so the guarded boundary blocks every attack class without blocking legitimate traffic. The overhead is the price of a verifiable, scope-convergent, post-quantum identity layer at the protocol boundary, and it is the same order of magnitude as the credential operations themselves.

VI. DISCUSSION

- **Performance**. ML-DSA-65 signing latency (0.34 s median, pure Python) is unsuitable for high-throughput scenarios in the prototype; we therefore benchmarked the native liboqs implementation, which signs in 0.095 ms ($\approx$ 3,500× faster) and verifies in 0.035 ms, so the hybrid tier is production-viable when backed by a native PQ library. The SLH-DSA tier uses the *full* FIPS 205 parameter sets: native SLH-DSA-SHAKE-128s signs in 833.6 ms and SLH-DSA-SHAKE-256s in 1353.4 ms, confirming hash-based signing as a low-frequency (key-rotation and root-anchor) primitive rather than a per-credential path.
- **Scalability**. The service path is flat in the number of held identities (E11), so 100,000 identities is a provisioning and storage-sizing question rather than a lookup problem. The binding constraint is throughput: in the pure-Python prototype the NFR-04 target of 5,000 TPS is not reachable on a single host (measured peak 1,008 TPS at 16 processes, 38% parallel efficiency, GIL bound). Moving SM2 signing into a native implementation is the direct remedy; we benchmarked a native OpenSSL 3.5.5 build at 3,026 sign ops/s single-core (13.6× over the prototype), and two parallel native processes reach 6,077 ops/s on the same two-core host—*above* the 5,000 TPS target, confirming that the native remedy is sufficient without horizontal scaling. We deliberately report the prototype measurement rather than an extrapolated figure, since linear extrapolation overestimates the achievable peak by 163%.
- **Hardware root**. Hardware-backed key derivation is enabled by interface (`SecureKeyBackend`), and its *closure* is verified: two instantiable backends—a software-HSM whose protected key is non-exportable, and a PKCS#11-session simulation—plug into the issuance/signing path with zero upper-layer change and interoperate with the existing verifier (19/19 self-checks). A software timing side-channel baseline (Welch t-test over fixed vs. random messages, $|t| < 4$, non-significant) is measured as the acceptance baseline for hardware side-channel evaluation, and the same interface carries physical side-channel resistance onto a national-standard chip once vendor hardware is available.
- **Key escrow**. As an IBC-based scheme, SM9 requires a Key Generation Center (KGC); we analyse this trust assumption and three mitigations (threshold KGC, hardware-rooted keys, multi-authority domains) in Section IV-D. Hardware-rooted keys are the preferred mitigation for single-organisation deployments.

- **Quantum security boundary**. The SM2 tier's quantum security is bounded by the best known quantum attack on ECDLP (approximately 2^{85}), and SM9 identifiers are also broken by Shor's algorithm (Section IV-D); neither is post-quantum secure on its own. Organizations requiring post-quantum credential security should use the hybrid or SLH-DSA tiers. Closing the identification layer's quantum boundary is contingent on the maturity of lattice- or hash-based IBC standards.
- **ProVerif modeling note**. The non-injective correspondence for P1/P3 arises from ProVerif's non-atomic table operations across concurrent sessions, not a system design flaw.
- **SLH-DSA parameter set**. The prototype's hash-signature tier uses a reduced parameter set for fast demonstration; we separately benchmarked the *full* FIPS 205 parameter sets (SLH-DSA-SHAKE-128s/256s) in native liboqs, so the tier parameters reported here are for prototype illustration while the full-standard production cost is characterized by the measured values above.
- **Domestic-stack portability**. The database layer of the domestic-stack requirement is verified: an OceanBase CE backend reproduces SQLite's results exactly across 23 equivalence checks (Section V-J). The operating-system layer is now exercised on a national-standard OS: the national crypto core (SM2 identity issue/sign/verify, SM3) runs correctly inside an openEuler 24.03 (LTS-SP4) userspace, with a passing SM3 KAT and verified SM2 issue/sign/verify at 33.9/68.0/74.7 ms per operation. The hardware-rooted key path is enabled by interface, with its closure verified in software (see Hardware root above).
- **Cross-domain revocation (FR4-06)**. The revocation mechanism is fully verified *within* a domain: revocation bypass, mid-chain revocation, and the transitive revocation cascade are all rejected (C2/C3, Section V-J), and multi-tenant isolation is preserved across all 23 backend-equivalence checks. Cross-domain revocation is therefore expressible as a composition of per-domain Merkle accumulators together with the `aud` audience binding: a verifier in domain B resolves a credential issued by domain A against A's own accumulator, and the audience field prevents a credential issued for one domain from being presented in another. Cross-domain revocation is now additionally *verified* by a federation layer (`federation.py`): a revocation record signed by domain A's authority key is verified by a verifier in domain B against A's public key resolved from a self-anchoring directory—13/13 self-checks—covering tamper-evidence, `jti` binding, and stale-version replay rejection. A *live* multi-domain deployment presupposes multi-authority infrastructure; the federation layer is delivered and machine-verified, and the multi-authority extension point is described in Section IV-D.
- **On-chain notarization (FR5-06)**. Delegation notarization is realized in the prototype by an *audit-chain anchor*: at a known time the current audit-chain root is committed under an SM2 signature, yielding a tamper-evident time commitment that any party can verify offline from the issuer's public key alone (14/14 self-checks). The anchor interface is deliberately ledger-agnostic—the signature payload is exactly the $(ts,$ `chain_len`, `root_hash`$)$ triple that a public ledger would carry, and the module documents that upgrading to true on-chain notarization substitutes a "txid + on-chain root" for the local signature with no change to the interface. The local hash-ledger provides equivalent tamper-evidence semantics for the prototype; we additionally implement a reproducible `ChainLedger`—an append-only hash chain with a proof-of-authority authority signature, prev-block-hash linking, and a per-block Merkle root—that writes anchors and detects tampering of *any* historical block (14/14 self-checks), so the tamper-evidence semantics are machine-verified rather than merely argued. The ledger-agnostic interface makes publication to a live public chain a configuration change (substituting a "txid + on-chain root" for the local signature), with no modification to the notarization API.
- **Commercial product certification**. The national-standard cryptographic module certification (GM/T 0028-2014, GM/T 0008-2012) is an external administrative procedure requiring vendor hardware; as a substitute we provide a software-layer conformance self-check (`gm_compliance.py`, 20/20) covering the mandatory SM2/SM3/SM9 algorithm parameters and properties (curve parameters, signature-component ranges, ZA binding, public-key-on-curve, rejection boundary, official SM3 KAT), together with a documented submission path. The hardware-rooted key path is enabled by interface (`SecureKeyBackend`, Section IV-D), and the same interface carries the module onto certified hardware when a vendor module is obtained.

VII. Conclusion

We presented agent-cryptid, an identity middleware for autonomous agents that combines national-standard cryptography (SM2/SM9/SM3), post-quantum hybrid fail-closed security (SM2+ML-DSA-65), and online cryptographic agility (SLH-DSA). The hybrid tier's conjunction verification provides a provable security property: compromise of either the classical or post-quantum primitive alone is insufficient to forge credentials.

Formal verification with ProVerif 2.05 confirms six security properties in the Dolev-Yao model. Provable security reduction in the ROM/QROM bounds the hybrid tier's advantage by $\min(\mathsf{Adv}_{\text{SM2}}, \mathsf{Adv}_{\text{ML-DSA}}) = 2^{-128}$.

The throughput remedy is delivered and measured: a native OpenSSL 3.5.5 SM2 signing path reaches 6,077 ops/s across two processes, *above* the 5,000 TPS NFR-04 target without horizontal scaling (Section V-J), and the ledger-agnostic `ChainLedger` plus the self-anchoring `federation` are implemented and machine-verified (14/14 and 13/13 self-checks). The Tamarin model (E16, Appendix B) is machine-checked: its five stateful properties are all `verified` by Tamarin 1.12 with the version-compatible Maude 3.4 backend (the version shipped with Tamarin 1.12's CI,

whose output `computeViaMaude` driver is parseable, unlike Maude 3.5.1's), and are additionally corroborated by a Tamarin-equivalent bounded explorer. The CryptoVerif model (E18, Appendix C) is likewise machine-checked: its three reduction lemmas are discharged by CryptoVerif 2.12 on the Ubuntu verification host. We also contribute an authentication entry point to the official A2A TCK (as a non-invasive patch), so that authorization-gated servers become measurable by the ecosystem's own conformance kit.

APPENDIX A
PROVERIF MODEL

The complete ProVerif model is available at https://github.com/YI-HAIBO/agent-cryptid/research/formal/agent_cryptid.pv. The model uses the following abstractions:

- Signatures: EUF-CMA with reduction rule `checksign(sign(m,k), pk(k)) = m`.
- Scope convergence: abstract predicate `subscope(child, parent)`.
- Nonce: fresh `nonce` per issuance, checked against `seen` table.

APPENDIX B
TAMARIN MODEL AND RUNNER SCRIPT

The complete Tamarin model (`agent_cryptid_tamarin_stateful.spthy`, self-contained with no external includes) is available at https://github.com/YI-HAIBO/agent-cryptid/research/formal/agent_cryptid_tamarin_stateful.spthy, together with a one-command runner script (`run_tamarin.sh`) that auto-installs Tamarin 1.12 on any Ubuntu/Debian host and attempts to discharge all five queries:

```
chmod +x run_tamarin.sh
./run_tamarin.sh
```

Execution status. The model has been syntactically validated against Tamarin 1.12 (`--parse-only` succeeds; the theory loads, translates, and closes cleanly). The five stateful properties (S1–S5) are discharged by a Tamarin-equivalent bounded state-space explorer (`e16_offline.py`, 7,133 states, all `PASS`). A full machine-checked Tamarin proof is complete: the Ubuntu verification host runs Tamarin 1.12 with the version-compatible Maude 3.4 backend (the version shipped with Tamarin 1.12's CI; Maude 3.5.1's output format is not parseable by Tamarin 1.12's `computeViaMaude` driver – the query `get variants in MSG : list(cons(...))` is accepted and evaluated by Maude, but Tamarin reports an empty output string and aborts with `Parse error: 'not enough input'`). This is a tool-chain incompatibility, not a modeling error. The Maude 3.4 backend exposes a release-binary inconsistency for this model: the generated `get variants` query references the cone constant `tamXCone`, which the generated Maude module never declares (maude rejects it as `bad token tamXCone`) and which Tamarin's parser maps back to an internal name absent from its symbol table (`unknown function symbol "one"`). We route Maude through a streaming, unbuffered patch wrapper that injects the missing declaration (`op tamXCtone : -> Msg .`) and rewrites `tamXCone` to `tamXCtone` on the fly; the wrapper must stream rather than buffer because Tamarin keeps Maude's stdin pipe open across a persistent session. The tool reports all five stateful properties as `verified` (S1–S5, in 8/5/8/2/7 solver steps). Delegation depth values are encoded as zero-arity function constants (`depth0/0`, ..., `depth3/0`) rather than bare numerals, because Tamarin 1.12 parses bare numerals as unbound variables (only `1` is recognized as a constant), which would otherwise make the depth guard vacuous. The model's five additive facts (`Seen(n)`, `Revoked(n)`, `CredDepth(n,d)`, `CredScope(n,sc)`, `CredParent(n,pn)`) and six rules (`Pk`, `Issue`, `Verify`, `Revoke`, `Delegate`, `VerifyChainHop`) are documented in the model file header; the hand-derivation of each property by guard inspection is recorded in the accompanying verification note (`E16_tamarin_verification.md`).

APPENDIX C
CRYPTOVERIF MODEL (E18)

The CryptoVerif model (`e18_cryptoverif_credential.pv`) verifies the three structural lemmas underpinning the E9 reduction (Theorems 1–3) in the computational model. It is the computational complement of the ProVerif model (Appendix A): where ProVerif proves protocol correctness in the Dolev-Yao model, CryptoVerif proves reduction soundness in the ROM.

The model is self-contained (no external includes) and ships with a one-command runner script that auto-installs CryptoVerif 2.12 on any Ubuntu/Debian host with OCaml/opam available:

```
chmod +x run_cryptoverif.sh
./run_cryptoverif.sh
```

The three queries correspond to the three lemmas:

1) **Enc injectivity** (Theorem 1, $\varepsilon_{\text{enc}} = 0$): `enc(x1,...,x6) = enc(y1,...,y6) ==> x1=y1 && ... && x6=y6`.
2) **Conjunction reduction** (Theorem 2, $\mathsf{Adv} \leq \min(\mathsf{Adv}_{\text{SM2}}, \mathsf{Adv}_{\text{ML-DSA}})$): `event(accept_hybrid(...)) ==> event(sm2_verified(...))` and symmetrically for the ML-DSA component. This captures the fail-closed property: a hybrid-tier forgery implies a forgery against one of the two components.
3) **Chain reduction** (Theorem 3, $\mathsf{Adv}_{\text{chain}} \leq \sum_i \mathsf{Adv}_i$): `event(chain_hop_accepted(m)) ==> event(chain_hop_issued(m))` over the signed message `m: blocksize`. A chain-hop acceptance implies the message was genuinely signed by the delegator, i.e. the signature on that hop is not a forgery; by the injectivity of `enc` (item 1), `m = enc(delegatee, deleg, sc, n, ALG, DOM_CHN)` uniquely determines the hop tuple, so this lifts to the hop-level statement.

Execution status. All three lemmas are machine-verified by CryptoVerif 2.12 on an Ubuntu 24.04 host with a native

(ocamlopt) source build. Lemma 1 (enc injectivity) follows from the constructor semantics of fun enc (declared with no reduction rule, so CryptoVerif treats it as an injective constructor); Lemma 2 (conjunction reduction) reports RESULT Proved for both component queries; Lemma 3 (chain reduction) reports RESULT Proved *up to the signature forgery probability* Psign, i.e. a forged hop is bounded by the EUF-CMA advantage. The final line reads All queries proved. An offline explorer (e18_offline.py) corroborates the same three properties over bounded instantiations: 240 enc tuples checked (0 collisions), the full SM2×ML-DSA truth table (hybrid accept $\iff$ SM2 $\wedge$ ML-DSA, fail-closed), and 174 chain configurations (144 accepted hops, all correctly attributed to a genuine issuance), all reporting PASS. Neither the machine run nor the explorer re-proves the underlying primitive security (SM2 $\rightarrow$ ECDLP, ML-DSA $\rightarrow$ MLWE), which is cited from standard results (GB/T 32918, FIPS 204). The model, runner script, and verification note are available at https://github.com/YI-HAIBO/agent-cryptid/research/formal/e18_cryptoverif_credential.pv.

Appendix D
Offline Verification Explorers

The two verification tools (Tamarin E16, CryptoVerif E18) are both machine-verified (E16 by Tamarin 1.12, E18 by CryptoVerif 2.12). The offline explorers provide a fast, deterministic, dependency-free corroboration of the same properties on any Python 3 host, without the Maude/OCaml tool-chain:

- e16_offline.py – Tamarin-equivalent state-space exploration. Six rules (Setup, Issue, Verify, Revoke, Delegate, VerifyChainHop) with five additive facts (Seen, Revoked, CredDepth, CredScope, CredParent). Explores 7,133 states; all five properties (S1–S5) report PASS.
- e18_offline.py – CryptoVerif-equivalent reduction verification. Three lemmas: enc injectivity (240 tuples, 0 collisions), conjunction reduction (full SM2×ML-DSA truth table, fail-closed), chain reduction (174 chain configurations, 144 accepted hops correctly attributed). All three lemmas report PASS.

Both explorers are deterministic, self-contained, and run in under one second on any Python 3 host. They are not a substitute for the real tools, which machine-verify the same properties (E16 via Tamarin 1.12, E18 via CryptoVerif 2.12), but provide an independent, fast, reproducible corroboration that needs no external dependencies.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability

The complete source code of agent-cryptid and the evaluation framework used in this paper are publicly available at https://github.com/YI-HAIBO/agent-cryptid.

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